OFFICIAL BRAND CATALOGUE • SENSORS

GEFRAN

Gefran S.p.A. (Italy / India)

Linear TWIIST® & Position Sensors, Resistance Thermometers (RTD), Melt Pressure & Drives

DEALER & SUPPLIER



ENGINEERING OVERVIEW & TECHNOLOGY HIGHLIGHTS

Gefran is a premier European manufacturer specializing in industrial position sensors (patented Linear TWIIST®, linear potentiometers, Hyperwave magnetostrictives, MEMS inclinometers, draw wires), precision Resistance Thermometers (RTD Pt100/Pt1000), industry-standard melt pressure transmitters for polymer processing, digital process controllers, and variable frequency drives.

- Patented Linear TWIIST® contactless multivariable transducers measuring position, speed, tilt, and vibration
- High-accuracy Resistance Thermometers (RTD): Pt100/Pt1000 Class A & 1/3 DIN, amplified 4-20mA (AR2/AR6), bayonet extrusion (TR5), and interchangeable MgO (TR2)
- Microprocessor controllers (Series 600, 800, 1200, 1600) and ADV200 high-performance vector inverters
- Complete Position Sensing: Linear potentiometers (LT/PK), magnetostrictive (WPG/MK4), rotary Hall sensors (GR3P), MEMS inclinometers (GIG), and draw-wire transducers (GSH)
- Industry-standard Melt Pressure Transducers for plastic extrusion up to 538°C (mercury, oil, NaK filled)
- Extreme measurement repeatability and durability in aggressive industrial environments

LINEAR TWIIST® CONTACTLESS POSITION TRANSDUCERS



LM-C Contactless Linear TWIIST Position Sensor (CANopen)

CANopen Multivariable

Contactless linear position transducer with TWIIST patented multi-variable technology. Measures position, velocity, triaxial tilt, vibration, and internal temperature over CANopen for Industry 4.0 preventive maintenance.

• Strokes: 100 to 1000 mm (1 µm resolution) • Multi-variable: Position, velocity, triaxial tilt, vibration & temp • Protocol: CANopen DS 301 / DSP 406 profile • Ingress & Temp: IP67 sealed, -40°C to +85°C, M12 5-pin



LM-L Contactless Linear TWIIST Position Sensor (IO-Link)

IO-Link Multivariable

Contactless multi-variable linear transducer featuring IO-Link v1.1 interface for smart factory integration, real-time condition monitoring, and edge diagnostics in industrial and mobile hydraulics.

• Strokes: 100 to 1000 mm (1 µm resolution) • Multi-variable: Position, speed, 3-axis inclination, vibration & temp • Protocol: IO-Link v1.1 (COM3 230.4 kBaud) & SIO mode • Ingress & Temp: IP67 sealed body, -40°C to +85°C, M12 5-pole



LS-A Contactless Linear TWIIST Position Sensor (Analog Output)

Analog Output

Patented contactless TWIIST position sensor with self-aligning mounting mechanism, available in single and redundant analog output configurations for machinery and mobile hydraulics.

• Strokes: 50 to 1000 mm • Outputs: 0-10 VDC, 4-20 mA, and ratiometric voltage • Configuration: Direct and reverse analog, single & redundant options • Mounting & Ingress: Self-aligning mechanism, IP67, -40°C to +85°C, M12

POSITION SENSORS (LINEAR, ROTARY, TILT & WIRE-DRAW)



Linear Potentiometric Transducers (LT / PK / PC Series)

Linear Potentiometer

High-precision resistive linear position transducers with conductive plastic tracks, delivering infinite resolution and excellent linearity for industrial automation.

• Strokes: 50 mm to 1500 mm • Resolution: Infinite, linearity up to ±0.05% • Design: Cylindrical rod and profile casing types • Protection: IP65 / IP67 sealed, temp up to +100°C



Magnetostrictive Linear Transducers (Hyperwave WPG / MK4)

Hyperwave Contactless

Non-contact absolute position transducers utilizing Hyperwave magnetostrictive technology for wear-free, ultra-long-distance measurement in tough conditions.

• Strokes: up to 4000 mm • Resolution: down to 0.5 µm with high update rate • Outputs: 0-10V, 4-20mA, CANopen, SSI, IO-Link • Cursor: Non-contact magnetic floating cursor or sliding cursor



Rotary Position & Angular Sensors (GR3P / GRA Series)

Contactless Rotary

Contactless Hall effect and conductive plastic angular displacement sensors designed for mobile machinery, steer-by-wire, and agricultural equipment.

• Measurement range: 0° to 360° continuous rotation • Technology: Magnetic Hall effect contactless sensing • Outputs: Single / dual redundant analog voltage, current, CANopen • Protection: IP67 / IP69K sealed for harsh outdoor environments



Industrial Inclinometers & Tilt Sensors (GIG / GIM Series)

MEMS Inclinometer

MEMS technology single and dual-axis inclinometers engineered for high-precision tilt angle measurement on aerial work platforms, cranes, and construction vehicles.

• Measurement range: ±10° up to ±180° / 360° • Accuracy: ±0.05° with thermal compensation • Outputs: 4-20mA, 0.5-4.5V, CANopen CiA 410 • Housing: Rugged zinc-plated alloy or plastic, IP67/IP69K



Draw Wire Displacement Sensors (GSH Series)

Wire-Draw Transducer

Compact wire-actuated position transducers that convert linear movement into rotary signals for long stroke displacement measurement on lifts and telescopic booms.

• Stroke: up to 10,000 mm (10 meters) • Wire: Highly flexible stainless steel measuring cable • Outputs: Potentiometric, 4-20mA, 0-10V, CANopen • Construction: Heavy-duty anodized aluminum enclosure

RESISTANCE THERMOMETERS (RTD PT100 / PT1000)



AR2 Series Amplified RTD with Interchangeable Insert

Amplified 4-20mA

Head-mounted resistance thermometer with interchangeable mineral insulated (MgO) insert and built-in 4-20mA loop-powered transmitter.

• Element: Pt100 Class A / 1/3 DIN according to IEC 751 • Output: 4-20 mA 2-wire amplified linear output • Temp range: -50°C to +400°C • Insert: Spring-loaded interchangeable MgO insert for easy maintenance



AR6 / AR6M Series Compact Amplified RTD Sensors

Compact Amplified

Compact resistance thermometers with integrated 4-20mA transmitter directly built into the sensor body for space-constrained industrial machinery.

• Stem: Stainless steel AISI 316 with mineral oxide MgO insulation • Output: 4-20mA (2-wire), factory calibrated • Electrical connection: M12 4-pin male connector or fixed cable • Protection: IP67 waterproof and vibration resistant



TR1 / TR1M Series General Purpose Head-Mounted RTDs

Head Mount RTD

Standard industrial resistance thermometers with aluminum connection head and fixed or mineral-insulated MgO stem for piping and vessels.

• Element: Pt100, Pt1000 (2, 3, or 4 wire connection) • Operating range: -200°C to +600°C • Connection head: DIN B, IP65 with ceramic terminal block • Mounting: Welded thread, compression fitting, or flanged thermowell



TR2 Series Interchangeable Insert RTD Sensors

Interchangeable MgO

Heavy-duty industrial RTD assemblies featuring spring-loaded interchangeable MgO inserts, allowing element replacement without draining process lines.

• **Construction: DIN 43772 compliant thermowell configuration** • **Element: Pt100 single or duplex (dual elements)** • **Shock & Vibration: Excellent resistance via compacted MgO insulation** • **Process connection: 1/2" Gas, 1/2" NPT, 3/4" NPT**



TR5 / TR5N Series Plastics Industry Bayonet RTDs

Plastics & Extrusion

Specialized resistance thermometers engineered with spring-loaded bayonet caps for temperature control in plastics extrusion, injection nozzles, and hot runners.

• **Fitting: Bayonet cap with adjustable tension spring** • **Cable: Metal braided stainless steel armor over Teflon insulation** • **Element: Pt100 DIN Class A with fast thermal response** • **Temp range: 0°C to +350°C**



TR6 / TR6M Series Flexible Cable-Type RTD Probes

Cable-Type Probe

Versatile cable-type resistance thermometers with mineral insulated MgO flexible sheath and bendable transition joint for tight machine spaces.

• **Sheath: AISI 316 bendable (bend radius 3x diameter)** • **Lead cable: Silicone, PTFE, or stainless steel overbraid** • **Stem diameters: 3 mm, 4.5 mm, 6 mm** • **Temp range: -50°C to +400°C / +600°C**

MELT PRESSURE & INDUSTRIAL PRESSURE TRANSDUCERS



Melt Pressure Transducers & Transmitters

Melt Pressure

Mercury-filled, FDA-compliant Oil-filled, and Sodium-Potassium (NaK) impact fluid-free melt pressure sensors for extruders.

• **Pressure range: 0-35 bar up to 0-2000 bar** • **Temp: up to 400°C / 538°C** • **Output: mV/V, 4-20mA, 0-10V, HART, CANopen, IO-Link**



Industrial Pressure Transducers

Pressure Transducer

Piezoresistive and bonded foil strain gauge pressure transmitters for hydraulic presses, pneumatic systems, and injection lines.

• **Gauge & absolute pressure 0-10 to 0-1000 bar** • **High shock & vibration resistance** • **Output: 4-20mA, 0-10V**

CONTROLLERS, INDICATORS & INDUSTRIAL DRIVES



Gefran Microprocessor Controllers (600, 800, 1200, 1600)

Series 600/800

Digital PID controllers with customizable bar-graphs, mathematical functions, timer alarms, and valve motor control.

• **Dual 4-digit LED or LCD display** • **Universal configurable input** • **RS485 Modbus RTU communication**



Gefran Inverters & Variable Frequency Drives (ADV200)

Field-oriented vector inverters engineered for extruders, cranes, metal processing, and high-speed precision spindles.

• **Power range: 0.75kW to 1.65MW** • **Sensorless and closed loop encoder feedback** • **Integrated safe torque off (STO)**

OFFICIAL COMMERCIAL QUOTATION & TECHNICAL SUPPORT

To request an instant pricing quotation, OEM warranty validation, or technical submittals for this brand:

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